Template

Studentnames and studentnumbers here

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Set-up your environment

Title Page

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Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The gender pay gap (GPG) remains an important social and economic issue in the European Union (EU). Although certain progress has been made toward improved gender equality, women still earn less than men on average in many European countries (Eurostat, 2024). Although the gender pay gap differs between countries and regions, differences in earnings between men and women still continue to be a problem across Europe.

Several factors lead to the gender pay gap. This includes things such as differences in working hours, segregation by occupation, career interruptions related to pregnancy and caregiving responsibilities, and unequal representation in higher-paying positions (Blau & Kahn, 2017). Recent research also reveals the fact that structural inequalities in labour markets and workplaces continue to influence earnings differences between men and women (International Labour Organization [ILO], 2024).

Understanding the gender pay gap is very important, since it affects women' s job opportunities, financial independence and also long-term economic security. Lower earnings throughout a woman's career can also lead to lower pension income and an increased risk of poverty later in life (European Commission, 2024). Also, wage inequality can have a negative impact on the economy because women may not always be able to fully use their skills and reach their full potential.

This study examines the gender pay gap across the EU by grouping Northern, Western, Southern, and Eastern European countries, and then comparing individual regions. Using wage, employment, and income data collected over multiple years, our research aims to identify regional patterns and trends in gender-based wage inequality.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

The gdp_data dataset summary:

The dataset "Gender pay gap in unadjusted form" comes from Eurostat. It measures the unadjusted gender pay gap, which is the difference in average gross hourly earnings between men and women. The dataset was last updated 3 months ago and contains 584 data cells.

- freq: Only one value is used, which is A (annual), so there is only one unique value.
- unit: Only PC (percentage) is used, so there is one unique value.
- nace_r2: Only the category [B-S_X_O] is included, so there is one unique value.
- geo: There are 39 geo codes instead of 41, meaning some geographic codes are missing from the extracted data. The missing codes seem to be: "LI" (Liechtenstein) and "EA21".
- TIME_PERIOD: The data covers 2007 to 2024.
- values: These represent the percentage difference in hourly wages between men and women.

The salary_data dataset summary:

The dataset "Average full time adjusted salary per employee" comes from Eurostat. This dataset shows the average salary per employee (full-time adjusted), allowing fair comparisons between countries regardless of differences in part-time work. The dataset was last updated 7 months ago and contains 1,560 data cells.

- freq: Only one value is used, A (annual), so there is one unique value.
- unit: Two units are used, EUR (euro) and NAC (national currency), so there are two unique values.
- geo: Includes 26 European countries and 2 additional aggregates, resulting in 28 unique geo entries. - TIME_PERIOD: The data covers the years from 1995 to 2024.
- values: Salaries are expressed in EUR and national currency (NAC).

```r
summary(gpg_data)
```

```
##         freq              unit             nace_r2               geo
##  Length    :584   Length    :584   Length    :584   Length    :584
##  N.unique :  1   N.unique :  1   N.unique :  1   N.unique : 39
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  1   Min.nchar:  2   Min.nchar:  7   Min.nchar:  2
##  Max.nchar:  1   Max.nchar:  2   Max.nchar:  7   Max.nchar:  9
##
##   TIME_PERIOD              values
##  Min.   :2007-01-01   Min.   :-1.30
##  1st Qu.:2011-01-01   1st Qu.: 9.70
##  Median :2016-01-01   Median :14.45
##  Mean   :2015-08-06   Mean   :13.87
```

```
##  3rd Qu.:2020-01-01   3rd Qu.:17.60
##  Max.   :2024-01-01   Max.   :30.90
```

```r
summary(salary_data)
```

```
##         freq             unit              geo          TIME_PERIOD
##  Length   :1560   Length   :1560   Length   :1560   Min.   :1995-01-01
##  N.unique :   1   N.unique :   2   N.unique :  28   1st Qu.:2004-01-01
##  N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2011-01-01
##  Min.nchar:   1   Min.nchar:   3   Min.nchar:   2   Mean   :2010-06-16
##  Max.nchar:   1   Max.nchar:   3   Max.nchar:   9   3rd Qu.:2018-01-01
##                                                     Max.   :2024-01-01
##      values
##  Min.   :    964
##  1st Qu.:  14585
##  Median :  25700
##  Mean   :  97584
##  3rd Qu.:  38114
##  Max.   :7297665
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Variables included in salary_data:

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: The unit of measure variable shows the measurement scale used for the data values. The unit of measures used in this dataset are EUR (euro) and NAC (National currency).
- geo: The variable geopolitical entity identifies 28 territorial units. It includes 26 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as two supranational aggregates.
- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 1995 to 2024. It is recorded in annual intervals using the YYYY format.
- values: the variable value is the annual salary in both euro (EUR) and the national currency (NAC).

Variables included in gpg_data:

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: shows the measurement scale used for the data values. The unit of measures used in this dataset is percentage (PC).

- nace_r2: The variable nace_r2 identifies the economic sector according to the NACE Rev. 2 classification. The category B-S excluding O includes industry, construction, and service activities, while excluding public administration, defence, and compulsory social security.
- geo: The variable geopolitical entity 41 territorial units. It includes 37 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as four supranational aggregates.
- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 2007 to 2024. It is recorded in annual intervals using the YYYY format.
- values: The variable value represents the unadjusted Gender Pay Gap (GPG), which is the percentage difference in average gross hourly earnings between male and female employees, relative to male earnings.

Part 3 - Quantifying

3.1 Data cleaning

In this section, we clean the Gender Pay Gap and salary datasets to prepare them for analysis. First, we check for missing values and remove incomplete observations, since missing data could affect the reliability of our results. Then, we keep only the variables needed for the analysis: country, year, and values.For the salary dataset, we keep only observations measured in euros (EUR) so that the salary values are comparable. We also remove aggregate regions such as EA19, EA20, and EU27_2020 because our analysis focuses on individual European countries. Finally, we rename the value variables and merge the two datasets by country and year.

3.2 Generate necessary variables

Here are descriptions for the three variables created in section 3.2:

Variable 1 – `region`

- This is a categorical variable, which assigns each EU country to of of the four European regions: Northern, Western, Southern or Eastern Europe.
- The classification we used follows the United Nations 'geo-scheme', and is aimed to uncover and show broader geographical patterns in the gender pay gap.
- By grouping EU countries into regions, we can compare the average pay gaps and salary levels across blocks which are both culturally and economically similar, rather than simply analysing individual countries in isolation.

Variable 2 – `gpg_severity`

- This is a continuous variable, which translates the relative gender pay gap (which expressed as a percentage in the data-set) into an absolute difference in annual earning in euros between women and men.
- It is calculated with the formula **`gpg_severity = (gpg / 100) * salary`**, where **`gpg`** is the unadjusted gender pay gap, expressed as a percentage, and **`salary`** is the average full-time adjusted salary per employee, expressed in Euros.
- Thus, the resulting value gives us the approximate amount, in euros, by which the average annual earnings of women are lower than that of men.

- The monetary severity this variable introduces to us, allows us to assess the economic impact on women across different European regions and associated years, which we would be unable to assess when only looking at the percentage gap indicating relative inequality.

Sub-variable – `region_summary` (aggregated dataset)

- This is not a variable on its own, it is rather a set of sub-variables relating to variable 2, which will make all of our graphical analysis possible from here on.
- For each region and year, this summary set computes the mean gender pay gap (`mean_gpg`), the mean salary (`mean_salary`), the mean gender pay gap severity (`mean_gpg_severity`), and the number of countries included (`n_countries`).
- These regional aggregates are thus used for the creation of the temporal and spatial variation visualisations further below in this report.
- They mainly smooth out noise which would be otherwise present if we plotted every country included, and reveal broader trends in each region over time.

Variable 1

Variable 2

Variable 3

3.3 Visualize temporal variation

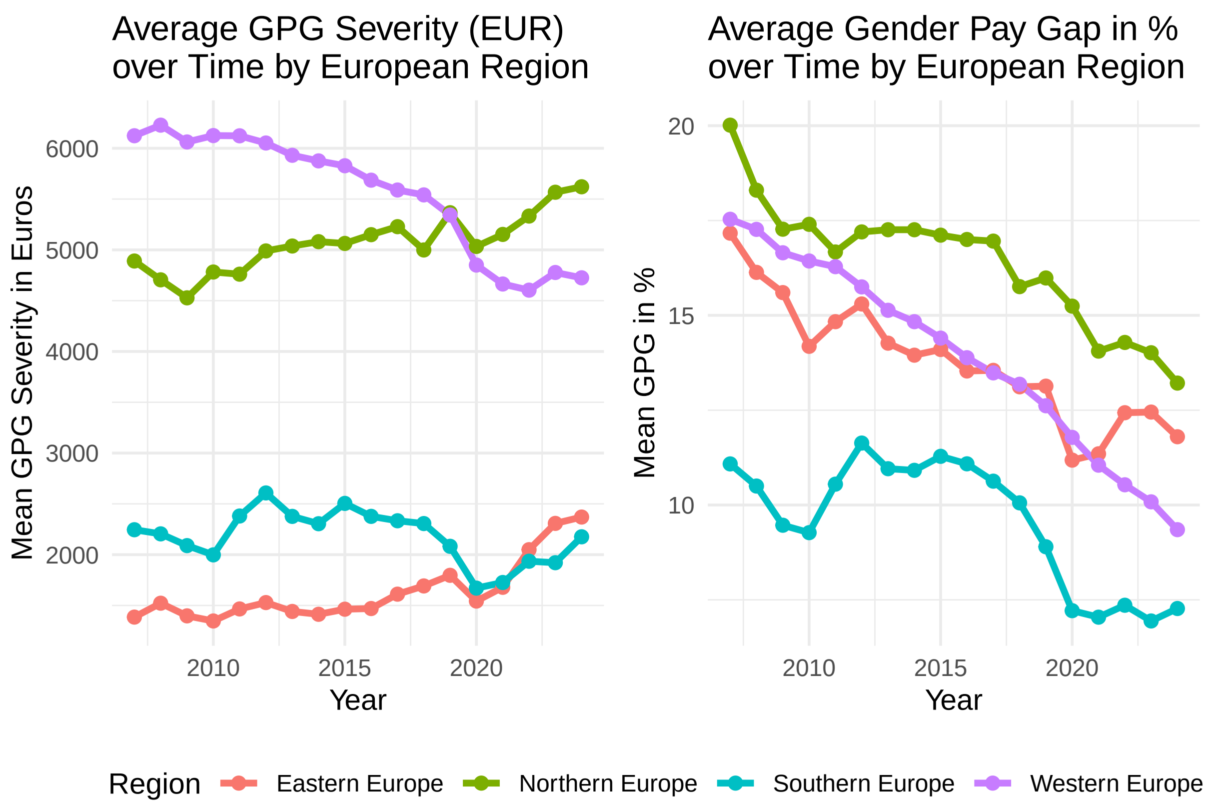
The graph for Visualisation of Temporal Variation consists of two line plots. The evolution of the Gender Pay Gap over time (x-axis) for the four European regions (coloured lines) is shown in both graphs.

The Left plot: Average Gender Pay Gap Severity in Euros over Time

- This plot shows the average/mean Gender Pay Gap Severity in Euros per Region per Year. GPG Severity, as we already discussed, shows the absolute gap in earnings between women relative to men, calculated with our formula: **`gpg_severity = (gpg / 100) * salary`**.
- When we express this gap in monetary terms, as can be seen in this plot, the real economic disadvantage women face is highlighted. A higher line therefore indicates a larger absolute gap in earnings. Further, this plot also allows us to see whether the impact of the pay gap has been growing, decreasing or remained constant over time across European regions.

The Right plot: Average Gender Pay Gap in % over Time

- This plot shows us the average unadjusted Gender Pay Gap as a percentage for the same European regions. When assessing the pay gap in terms of percentages, it makes it easier for us to compare and see the relative inequality across the four European regions.
- A declining line therefore suggests that the relative difference between women's and men's earnings is narrowing. We can also observe that despite different wage levels, such as between Eastern and Western Europe, these two regions have a similar pay gap relative to the associated incomes in those regions.



3.4 Visualisation of Spatial Variation

The figure for Visualisation of Spatial Variation consists of two Choropleth maps showcasing all EU countries, both of which are using the most recent year of data available from our merged dataset. The maps' colour gradient ranges from yellow (low values) to green (middle values) to purple (high values).

The Left map: Gender Pay Gap Severity in Euros

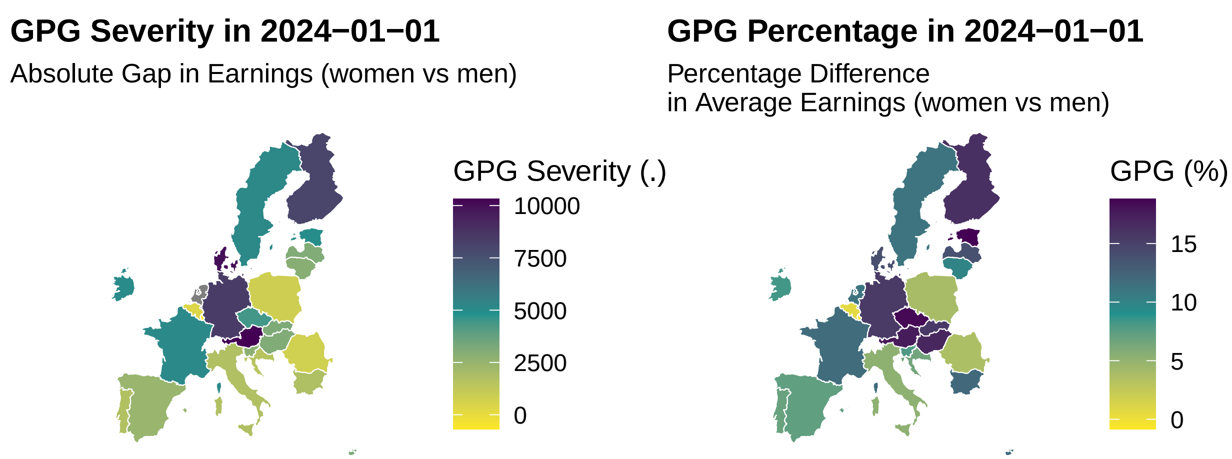
- The fill colour in this map shows the Gender Pay Gap Severity in Euros for each EU country. Darker purple indicates a larger absolute gap in earnings for women.
- The map also directly shows us which countries are the main culprits behind this problem - i.e. which countries' women experience the heaviest financial and economical penalty as a result of the Gender Pay Gap. Even if the percentage gap in those countries appears moderate (e.g. Sweden, which is a high salary economy), the severity is indeed still high in monetary terms.

The Right map: Gender Pay Gap in %

- The fill colour in this map shows the unadjusted Gender Pay Gap as a percentage for each EU country.
- This map, on the other hand, reflects the relative inequality of how much less women earn compared to men, for each country. This map also helps us to identify EU countries with the most unequal wage structure - regardless of the general wage level. We can also notice that some countries have both a high GPG Severity and a high GPG in %: Germany, Austria, Finland and Denmark are on the top of such list.

Relevance to the social problem

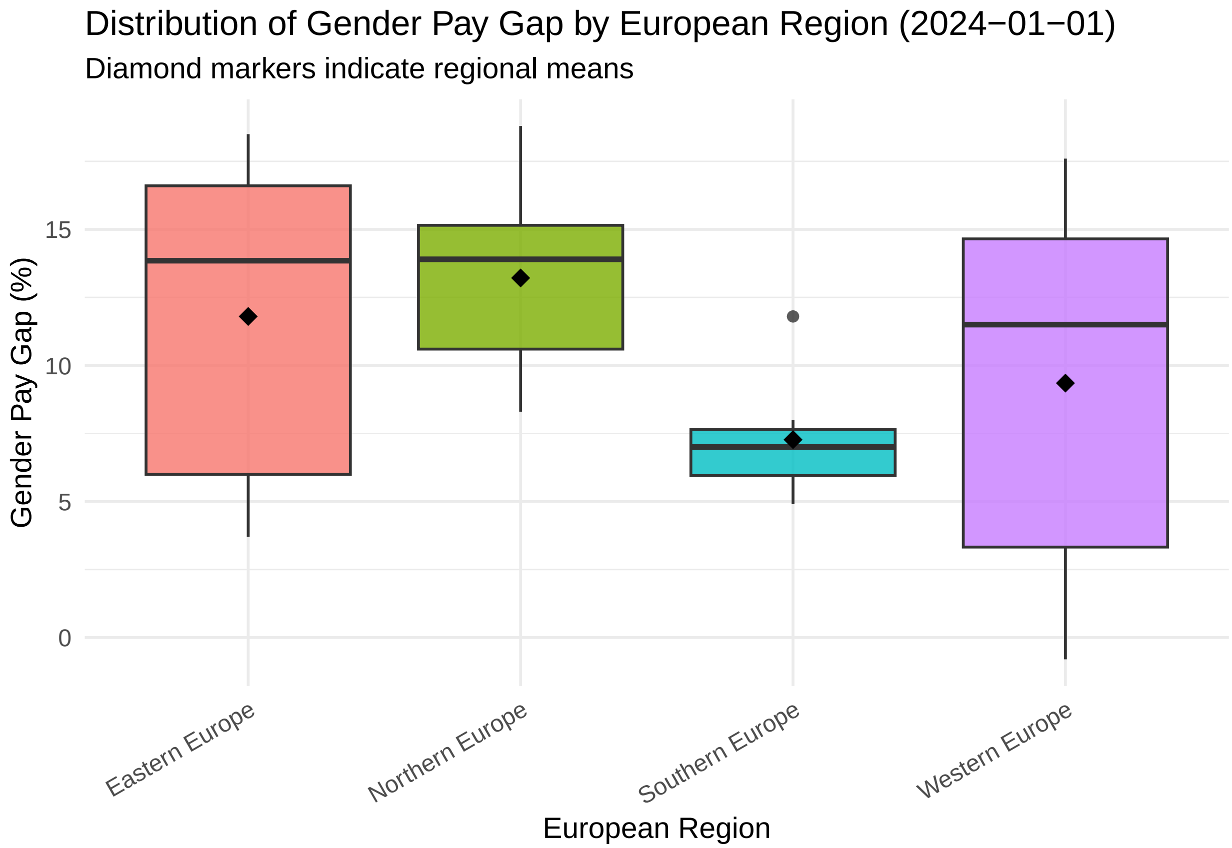
- This Spatial Variation analysis reveals rather striking regional patterns across the EU.
- Let's re-examine an aforementioned example: Eastern European countries often display lower GPG Severity compared to Western Europe, however, when looking at the GPG in %, we can see that, for example, countries such as Slovakia and Czechia are at similarly severe level as Germany and Austria in terms of percentage gaps.
- The Western European region has both high GPG Severity and high GPG in %, as average salaries are higher in that given region, while conversely, Southern Europe is in general the most equal amongst our examined regions, with having both the lowest GPG Severity and lowest GPG in %.
- By comparing the two maps, we can see that the Gender Pay Gap is indeed not uniform across the EU, and that effective policies must certainly take into consideration the relative gap (GPG in %) experienced and the absolute economic harm (GPG Severity) done to women. Furthermore, our maps enable quick identification of 'GPG Hotspots', which require more urgent attention.



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

The boxplot illustrates the distribution of the Gender Pay Gap across four European regions in 2024.By presenting both the median and mean values for each region, the figure provides a clearer basis for comparing regional patterns of gender pay inequality across Europe.



It is clearly shown that Northern Europe has the highest median Gender Pay Gap which clearly shows that most of the countries recording value above 10%. By contrast, Southern Europe has the lowest Gender Pay Gap and it also shows the smallest variation between countries. According to the box plot about Western Europe, it indicates relatively high Gender Pay Gap values, although some lower observations reduce the regional mean. The greatest variation is seen in Western Europe, indicating that gender pay inequality varies greatly among the nations in this region.

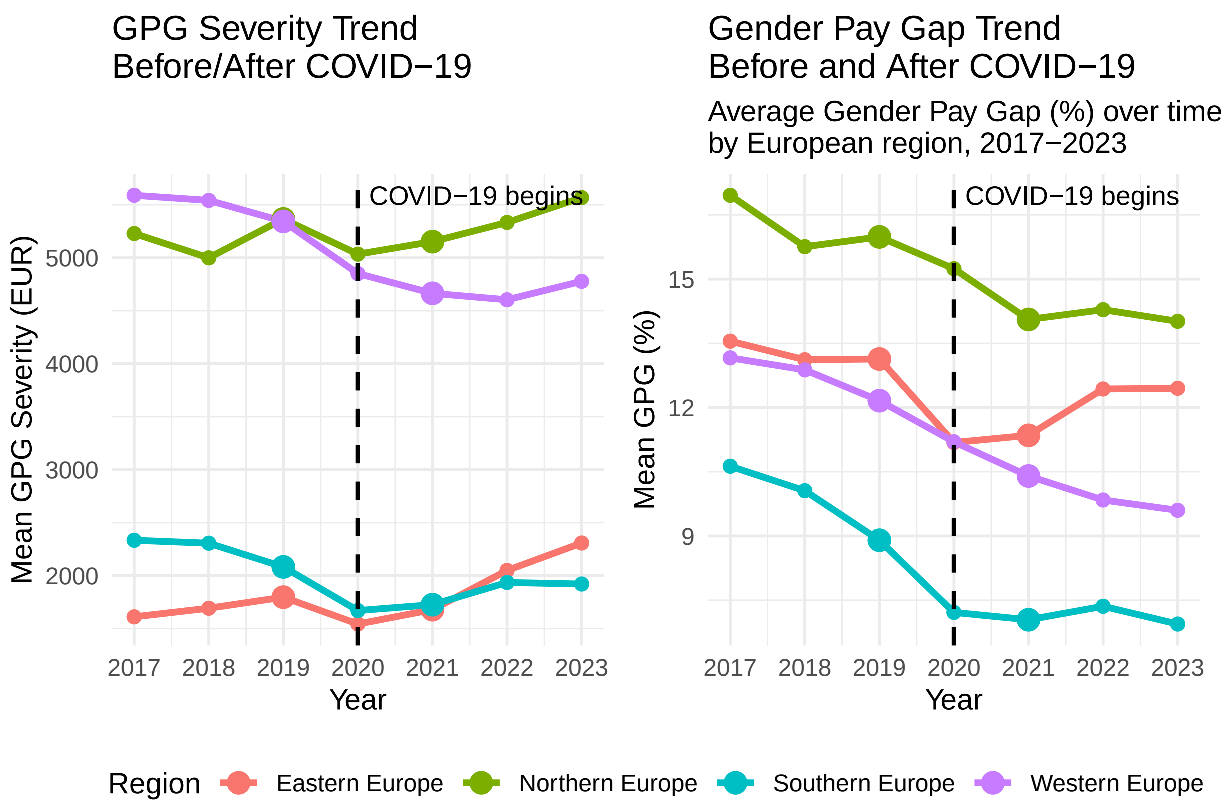
In conclusion, these results show that the Gender Pay Gap is not evenly distributed across Europe, which makes the analysis relevant to the social issue of gender pay inequality. Regional differences may reflect variations in labour market structures, economic conditions, and social policies. In addition, the wide range observed in Western Europe suggests that regional averages may not fully represent the experiences of individual countries

3.6 Event analysis

This event analysis examines the Gender Pay Gap before and after the COVID-19 . The dashed vertical line represents the year 2020, when COVID-19 began to significantly affect European labour markets. Comparing the trends before and after this event helps us understand whether the pandemic was associated with changes in gender pay inequality

```
## Dataset query already saved in cache_list.json...

## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/eurostat/5afdb14a965

## Table  tesem180  read from cache file:  /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/
```

```
## Dataset query already saved in cache_list.json...

## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5TxGN/eurostat/ebe94ad10cb

## Table  nama_10_fte  read from cache file:  /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//Rtmpn5T
```



The line graphs indicate that Southern Europe performs relatively well in general. Throughout the study period, it consistently records the lowest Gender Pay Gap percentages and the lowest GPG Severity values. This suggests that both the relative pay gap and the actual income difference between men and women are smaller in this region.

In comparison, Northern Europe clearly shows the inequality in payment between the woman and the man. Although the Gender Pay Gap percentage decreased slightly since 2020, GPG severity remains high and increased after Covid-19. Although the relative pay gap has narrowed slightly, the actual income gap between men and women remains quite significant

Western Europe shows the most noticeable improvement after COVID-19. Both the Gender Pay Gap percentage and euro-based GPG Severity decline over time. It is indicate that reduction in gender pay inequality in both relative and absolute terms. Eastern Europe presents a more complex pattern. Although the Gender Pay Gap percentage decreases, the actual salary gap continues to widen

Part 4 - Discussion

4.1 Interpretation of Findings One of the more striking results is that some of Europe's wealthiest regions still face real financial consequences from the gender wage gap. Western and Northern European countries usually score well on broader equality measures, but because their average salaries are so high, even a modest percentage gap can add up to a big difference in actual earnings. This lines up with earlier research arguing that prosperity alone doesn't guarantee fair pay. Inequalities can stick around even in highly developed economies (European Commission, 2024). Part of the explanation likely comes down to occupational segregation. Across Europe, men and women are still spread unevenly across different sectors. Better-paying fields like finance, tech, and engineering remain male-dominated, while women are overrepresented in areas such as education, healthcare, and social services. As a result, pay gaps can linger even where strong equal-pay laws are in place. These patterns don't mean that a region's location itself causes the pay gap. What we're seeing more likely reflects a mix of institutional, economic, demographic, and cultural factors—things like childcare access, parental leave, education levels, labour market flexibility, and social norms. So these results are best read as associations, not proof of cause and effect.

4.2 Limitations Several limitations should be considered when interpreting these findings. First, the study uses Eurostat's unadjusted gender pay gap measure, which does not account for factors such as education, occupation, work experience, or hours worked. As a result, the analysis cannot determine whether observed wage differences are driven by discrimination or other labour market characteristics. Second, the data is aggregated at the country level. While this helps identify broad regional patterns, it may mask important differences between individuals, industries, and firms within countries. In addition, grouping countries into broad regions simplifies comparison but may overlook significant variation between countries with different labour market structures, welfare systems, and social norms. Finally, this study identifies patterns and associations rather than causal relationships. Although regional differences are observed, other unmeasured factors may influence both salary levels and gender pay gap outcomes. Therefore, the results should be interpreted with caution.

4.3 Suggestions for Future Research Future research could build on this study by exploring additional factors that may influence the gender pay gap, such as education levels, labour force participation, childcare availability, and parental leave policies. Examining individual countries rather than broad regions could also provide a more detailed understanding of national differences. In addition, comparing adjusted and unadjusted gender pay gap measures may help distinguish between labour market characteristics and unexplained wage disparities. Finally, using more advanced statistical methods could provide stronger evidence on whether specific policies contribute to reducing gender wage inequality. Overall, further research is needed to better understand the factors that continue to shape gender pay differences across Europe.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.

include=FALSE -> to be used when we want to hide code